A Summer Industry Internship –II
Report on

CYBER SECURITY

During
IV Year I Semester Summer

Submitted to
The Department of Electronics and Communication engineering
In partial fulfilment of the academic requirements of
Jawaharlal Nehru Technological University
For
The award of the degree of

Bachelor of Technology in
ELECTRONICS AND COMMUNICATION ENGINEERING
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CERTIFICATE

This is to certify that this Summer Industry Internship – I Report on Internship carried out in "**Cybersecurity**", submitted by **MD.MUBEEN AHMED (21311A04BA)** in the year 2024in partial fulfilment of the academic requirements of Jawaharlal Nehru Technological University for the award of the degree of Bachelor of Technology in Information Technology, is a bonafide work in summer industry internship that has been carried out **during IV B Tech ECE I semester, will be evaluated in IVB Tech ECE I Semester**. This report has not been submitted to any other institute or university for the award of any degree.

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INTERNSHIP CERTIFICATE

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During April - June 2024

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INTERNSHIP GRADE P

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

DECLARATION

I MD.MUBEEN AHMED (21311A04BA) student of **SREENIDHI INSTITUTE OF SCIENCE AND**
TECHNOLOGY, YAMNAMPET, GHATKESAR, studying IV year Ist semester,
ELECTRONIC AND COMMUNICATION ENGINEERING solemnly declare that the Summer Industry Internship-II Report, is submitted to **SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY** for partial fulfillment for the award of degree of Bachelor of technology in **ELECTRONIC AND COMMUNICATION ENGINEERING** .

It is declared to the best of our knowledge that the work reported does not form part of any dissertation submitted to any other University or Institute for award of any degree

MD.MUBEEN AHMED

21311A04BA

ACKNOWLEDGEMENT

I would like to express my gratitude to all the people behind the screen who helped me to transform an idea into a real application.

I would like to thank my Project coordinator **Mrs.K.Vineetha** for her technical guidance, constant encouragement and support in carrying out my project at college.

I profoundly thank **Dr. Subba Rao** Head of the Department of electronics and communication engineering who has been an excellent guide and also a great source of inspiration to my work.

I would like to express my heart-felt gratitude to my parents without whom I would not have been privileged to achieve and fulfill my dreams. I am grateful to our principal, **Dr. T. Ch. Siva Reddy,** who most ably run the institution and has had the major hand in enabling me to do my project.

The satisfaction and euphoria that accompany the successful completion of the task would be great but incomplete without the mention of the people who made it possible with their constant guidance and encouragement crowns all the efforts with success. In this context, I would like thank all the other staff members, both teaching and non-teaching, who have extended their timely help and eased my task.

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1.Introduction

1.1 About the internship:

The Forge certification is the first of its kind credential to cover foundational knowledge of industry recognized cybersecurity and network security concepts as well as various cutting-edge advancements across all Palo Alto Networks technologies. As the cybersecurity landscape becomes more complex, Palo Alto Networks Education Services has taken steps to align with industry standards following the NIST/NICE (National Institute of Standards and Technology/National Initiative for Cybersecurity Education) workforce framework. This cybersecurity is a part of Forge internship program, there were streams like AIML, AWS, android web development, cloud computing etc. The cybersecurity is the stream I opted and managed by Pala Alto Networks.

1.2 Plan of training program:

The program was held for 10 weeks starting from 1st of April 2024 and was ending on 30 th June 2024. In this course we had 4 topics to be covered and each topic have different modules and each module have lessons based on concepts to be covered in that module. In each module ending we had assessment based on knowledge gained. If the assessment is qualified then the next module available. At last, we had to write the total exam based on all the four topics if only 85% percent then the final certification is achieved.

1.3 Scope:

With the current spike in demand for cybersecurity specialists, there are more cybersecurity jobs on the market than qualified candidates. Cybersecurity employment will be in great demand in the future days if the current trend continues. India's cybersecurity colleges have been offering a variety of courses that are relevant to the shifting conditions.

Cybersecurity is a burgeoning career field. As the global business environment transitions to cloud data storage and online administration, demand for cybersecurity. Commercial organization data and use personal data are at risk of being misused as the internet becomes more widely used. This has boosted the demand for cybersecurity experts who are conversant with and skilled in the field.

The ever-changing technical landscape needs recruiting brilliant people with varying degrees of knowledge, which is one of the key reasons for the industry's rapid growth. While there are many **job openings in cybersecurity**, suitable applicants are in short supply, as this area demands specialized knowledge that is typically taught in a cyber security professional degree and training programs.

1.4 Technology learnt:

Cybersecurity technologies in simpler terms:

1. Firewalls: Think of firewalls like a security guard for your computer or network. They monitor and control the incoming and outgoing traffic to keep out any unauthorized or potentially harmful stuff.

2. Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS): These are like security alarms that detect and prevent any suspicious activities or attempts to break into your computer or network.

3. Encryption: It's like putting your data in a secret code. Encryption makes your information unreadable to anyone who doesn't have the key, ensuring that only authorized people can access and understand it.

4. Authentication and Access Control: These are like locks and keys. They verify your identity and control who can access certain parts of a system or data, making sure only the right people can get in.

5. Security Information and Event Management (SIEM): It's like having a security team that collects and analyzes information from different sources to detect and respond to potential security threats.

6. Vulnerability Assessment and Penetration Testing (VAPT): These techniques help find weaknesses in systems by testing them for potential vulnerabilities, just like a simulated attack, so that organizations can fix those weaknesses before real attackers find them.

2. Introduction to cyber security

2.1 Introduction to cyber security:

Cybersecurity, short for "cybersecurity," is a field dedicated to protecting computer systems, networks, and data from unauthorized access, attacks, damage, or theft. As our world becomes increasingly interconnected through digital technology, the importance of cybersecurity has grown exponentially. The primary goal of cybersecurity is to ensure the confidentiality, integrity, and availability of information in the digital realm.

Key Concepts in Cybersecurity:

Confidentiality: Protecting sensitive information from unauthorized access. This involves implementing measures such as encryption and access controls to ensure that only authorized individuals or systems can view certain data.

Integrity: Ensuring the accuracy and reliability of data. Cybersecurity measures prevent unauthorized alterations to information, assuring that data remains intact and trustworthy.

Availability: Ensuring that information and systems are accessible when needed. Cybersecurity strategies aim to prevent or mitigate disruptions to services, whether due to technical failures or malicious attacks.

Authentication: Verifying the identity of users, systems, or devices. Authentication methods include passwords, biometrics, and multi-factor authentication to ensure that only authorized entities gain access.

Authorization: Granting appropriate permissions and access levels to users or systems based on their authenticated identity. This helps prevent unauthorized access to sensitive information.

Network Security: Protecting the integrity and confidentiality of data as it is transmitted over networks. This involves measures such as firewalls, intrusion detection and prevention systems, and virtual private networks (VPNs).

Endpoint Security: Securing individual devices (endpoints) such as computers, smartphones, and tablets. This includes antivirus software, anti-malware programs, and device encryption.

Incident Response: Planning and responding to cybersecurity incidents. This involves identifying and mitigating the impact of security breaches, as well as learning from these incidents to improve future response strategies.

Security Policies and Procedures: Establishing guidelines and rules to govern the behavior of users and systems within an organization. This includes educating users on best practices and ensuring compliance with security policies.

Security Awareness Training: Educating individuals about potential cybersecurity threats and how to protect themselves and their organizations. This is crucial for building a securityconscious culture.

The constantly evolving nature of cyber threats requires cybersecurity professionals to stay informed about the latest developments and adopt proactive measures to safeguard against emerging risks. As technology continues to advance, the importance of cybersecurity in safeguarding digital assets and ensuring the smooth functioning of our interconnected world will only continue to grow.

Delving deeper into cybersecurity involves exploring advanced concepts and technologies designed to address the ever-changing landscape of cyber threats. Here are some deeper aspects of cybersecurity:

Cryptography: At the heart of many cybersecurity measures, cryptography involves the use of mathematical algorithms to secure communication and protect information. This includes encryption (converting data into a secure code) and hashing (creating a fixed-size output from any input).

Penetration Testing (Ethical Hacking): Ethical hackers, or penetration testers, simulate cyber attacks to identify vulnerabilities in systems, networks, or applications. This proactive approach helps organizations identify and address potential weaknesses before malicious actors can exploit them.

Security Information and Event Management (SIEM): SIEM systems collect and analyze log data from various sources within an organization's IT infrastructure. By correlating this information, SIEM tools can detect and respond to security incidents in real-time.

Artificial Intelligence (AI) and Machine Learning (ML): AI and ML play an increasingly important role in cybersecurity. These technologies can analyze vast amounts of data to identify patterns, detect anomalies, and improve threat detection capabilities. They are used in antivirus programs, intrusion detection systems, and behavioral analysis tools.

Zero Trust Security Model: This model assumes that threats can come from both outside and inside the network. It advocates for continuous verification of the identity and security posture of every device and user attempting to access resources, regardless of their location.

Blockchain Security: As blockchain technology gains prominence, securing decentralized systems becomes crucial. Blockchain's inherent characteristics, such as decentralization and immutability, can be leveraged to enhance the security of various applications, including cryptocurrencies and smart contracts.

Cloud Security: With the widespread adoption of cloud computing, ensuring the security of data stored in the cloud is paramount. Cloud security involves measures such as encryption, access controls, and secure APIs to protect information and applications hosted in the cloud.

Threat Intelligence: Gathering and analyzing information about potential cybersecurity threats is essential for proactive defense. Threat intelligence involves monitoring the global threat landscape to identify emerging risks, vulnerabilities, and attack patterns.

Security Automation and Orchestration: Automating repetitive security tasks and orchestrating complex workflows can enhance the efficiency of cybersecurity operations. This includes automating incident response, threat detection, and patch management processes.

Quantum Computing Threats and Post-Quantum Cryptography: As quantum computing advances, it poses a potential threat to current cryptographic methods. Post-quantum cryptography aims to develop algorithms that can resist attacks by quantum computers, ensuring the long-term security of sensitive information.

Deepening one's understanding of cybersecurity involves staying abreast of technological advancements, emerging threats, and evolving best practices. Professionals in this field often pursue certifications, attend conferences, and participate in ongoing training to keep their skills and knowledge up-to-date in the face of an ever-evolving cyber landscape.

2.1 Types of Cyber security:

Cybersecurity is a multifaceted field with various specialized areas of focus. Different types of cybersecurity can be categorized based on the nature of the threats they address and the areas of protection they emphasize. Here are some common types of cybersecurity:

Network Security:

Firewalls: Implementing hardware or software barriers to control incoming and outgoing network traffic, acting as a barrier between a trusted internal network and untrusted external networks.

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS): Monitoring network and/or system activities for malicious activities or security policy violations and taking preventive actions.

Endpoint Security:

Antivirus and Anti-Malware: Protecting individual devices (endpoints) from malicious software and other security threats.

Device Encryption: Securing data on individual devices through encryption, preventing unauthorized access in case the device is lost or stolen.

Application Security:

Web Application Security: Focusing on securing web applications from security threats, including injection attacks, cross-site scripting (XSS), and other vulnerabilities.

Mobile Application Security: Protecting mobile apps from security threats and vulnerabilities that could be exploited to compromise data or the device.

Cloud Security:

Cloud Access Security Broker (CASB): Providing security policies and enforcement for data hosted on cloud services.

Identity and Access Management (IAM): Managing and controlling user access to cloud resources to ensure proper authentication and authorization.

Data Security:

Encryption: Protecting sensitive data by converting it into a secure format that can only be decrypted with the appropriate key.

Data Loss Prevention (DLP): Implementing measures to prevent unauthorized access, sharing, or loss of sensitive data.

Identity and Access Management (IAM):

Authentication: Verifying the identity of users through methods like passwords, biometrics, and multi-factor authentication.

Authorization: Controlling access rights and permissions based on the authenticated identity of users.

Incident Response and Management:

Incident Response Planning: Developing strategies and processes to identify, respond to, and recover from cybersecurity incidents.

Forensics: Investigating and analyzing cybersecurity incidents to understand the nature of the attack and gather evidence.

Security Awareness and Training:

Educating Users: Training individuals within an organization to recognize and avoid security threats, phishing attacks, and other potential risks.

Security Policies: Establishing and enforcing policies that guide user behaviour and actions to maintain a secure environment. Internet of Things (IoT) Security:

Securing IoT Devices: Implementing security measures for connected devices to prevent unauthorized access and protect data integrity.

Network Segmentation: Isolating IoT devices from critical systems to contain potential security breaches.

Critical Infrastructure Security:

Industrial Control Systems (ICS) Security: Protecting systems that control critical infrastructure such as power plants, water treatment facilities, and transportation systems from cyber threats.

SCADA (Supervisory Control and Data Acquisition) Security: Ensuring the security of systems that monitor and control industrial processes.

These types of cybersecurity measures often overlap, and a comprehensive cybersecurity strategy involves a combination of these approaches to create a robust defense against a wide range of threats. The choice of specific cybersecurity measures depends on the organization's needs, the nature of its assets, and the potential risks it faces.Top of Form

Cybersecurity is a broad and critical field that focuses on protecting computer systems, networks, and data from various forms of cyber threats. The increasing reliance on digital technologies and the interconnected nature of modern systems make cybersecurity a crucial aspect of both individual and organizational safety. Here's a detailed overview of various aspects of cybersecurity:

2.3 Cyber Threats:

Malware: Malicious software, such as viruses, worms, Trojans, ransomware, and spyware, designed to harm or exploit systems.

Phishing: Deceptive attempts to obtain sensitive information, often through fraudulent emails, websites, or messages.

Denial of Service (DoS) and Distributed Denial of Service (DDoS): Overloading a system or network to disrupt its normal functioning.

Man-in-the-Middle Attacks: Intercepting and manipulating communication between two parties without their knowledge.

Cybersecurity Measures:

Firewalls: Network security devices that monitor and control incoming and outgoing network traffic based on predetermined security rules.

Antivirus Software: Programs designed to detect, prevent, and remove malicious software.

Encryption: The process of converting data into a code to prevent unauthorized access.

Multi-Factor Authentication (MFA): Adding an extra layer of security by requiring multiple forms of identification.

Network Security:

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS): Monitoring network or system activities for malicious activities or security policy violations.

Virtual Private Networks (VPNs): Securely connecting remote users or networks to a private network over the internet.

Security Protocols: Implementing secure communication protocols such as SSL/TLS for data encryption.

Application Security:

Secure Coding Practices: Writing code with security considerations to prevent vulnerabilities.

Web Application Firewalls (WAF): Protecting web applications from various attacks and vulnerabilities.

Regular Software Updates and Patch Management: Keeping applications and systems up-todate to address known security flaws.

Incident Response and Forensics:

Incident Response Plans: Developing and implementing plans to respond to and recover from cybersecurity incidents.

Digital Forensics: Investigating and analyzing digital evidence to determine the cause and impact of a cybersecurity incident.

Security Policies and Compliance:

Security Policies: Establishing rules and guidelines to ensure the confidentiality, integrity, and availability of information.

Compliance Frameworks: Adhering to industry or regulatory standards to protect sensitive data.

Security Awareness and Training:

Employee Training: Educating users about cybersecurity best practices and potential risks.

Security Awareness Programs: Building a culture of security within organizations.

Emerging Technologies in Cybersecurity:

Artificial Intelligence (AI) and Machine Learning (ML): Enhancing threat detection and response capabilities.

Blockchain Technology: Securing transactions and data through decentralized and tamperresistant ledgers.

Global Cybersecurity Challenges:

White Hat Hacking: Ethical hackers testing systems for vulnerabilities to improve security.

Penetration Testing: Simulating cyber attacks to identify and address weaknesses.

Cybersecurity is a dynamic field that constantly evolves as new technologies and threats emerge. Staying informed, adopting best practices, and implementing a multi-layered security approach are essential for effectively mitigating cyber risks.

3. Internship feedback.

3.1 About company

When it comes to feedback on companies like Palo Alto Networks in the cybersecurity industry, it's important to consider multiple perspectives and sources. Palo Alto Networks is a well-known company that offers a range of cybersecurity solutions, including firewalls, threat intelligence, and cloud security. They have a good reputation for their advanced security technologies and commitment to protecting organizations from cyber threats. However, it's always a good idea to do your own research and read reviews to get a comprehensive understanding of their products and services.

Palo Alto Networks are a leading cybersecurity company known for their advanced security solutions. Palo Alto Networks offers a wide range of products and services, including nextgeneration firewalls, threat intelligence, endpoint protection, and cloud security. Their solutions are designed to help organizations protect their networks, data, and applications from various cyber threats such as malware, ransomware, and advanced persistent threats. Palo Alto Networks has a strong reputation in the industry and is trusted by many organizations worldwide.

3.2 Experience in Internship:

It was really a great experience in learning all security concepts well in advance. The material provided by Pala alto networks was really a meaningful information one. The interest in learning things was increasing as course goes on. Learnt many techniques like encryption, network services, decryption, cyber attacks and many more. The interestingness was increasing as there were assessments conducted in the form of tests and thereby testing how far we got the topic read previously.

The course had four topics to be covered with different areas to be covered

1.Introduction to cyber security

2.Cloud security fundamentals

3.Network security fundamentals

4.Security operation fundamentals

With all these being covered , I understood how the data is organized, how the data attacks are taking place, how the data being stored, how the data is transferred with networks topologies and protocols etc, and how the data being protected and sent by sender and being received by receiver which is to be decrypted which include concepts like Kerberos, digital signature, RSA algorithms etc that provides security to data.

3.3 Challenges faced:

In the field of cybersecurity, there are several challenges that organizations and individuals face. One common challenge is the constantly evolving nature of cyber threats. Hackers and cybercriminals are always finding new ways to exploit vulnerabilities and launch sophisticated attacks. This requires cybersecurity professionals to stay updated with the latest trends and technologies to effectively defend against these threats.

Another challenge is the shortage of skilled cybersecurity professionals. The demand for cybersecurity expertise is high, but there is a shortage of qualified professionals to fill these roles. This scarcity makes it difficult for organizations to build strong cybersecurity teams and puts them at risk of potential attacks.

Additionally, the complexity of modern IT environments poses a challenge. With the increasing use of cloud computing, IoT devices, and interconnected networks, securing all aspects of an organization's infrastructure becomes more complex. It requires comprehensive strategies and solutions to ensure the protection of data and systems across various platforms.

Lastly, user awareness and behavior can also be a challenge. Human error, such as falling for phishing scams or using weak passwords, can expose organizations to cyber threats. Educating users about best practices and promoting a culture of cybersecurity awareness is crucial in mitigating these risks.

These are just a few of the challenges faced in cybersecurity, and addressing them requires a combination of technical solutions, skilled professionals, and a proactive approach to security.

4. Fundamentals of Cloud Security

4.1 Cloud Computing

The move toward cloud computing not only brings cost and operational benefits but also technology benefits. Data and applications are easily accessed by users no matter where they reside, projects can scale easily, and consumption can be tracked effectively.

One value of cloud computing is the ability to pool resources to achieve economies of scale. This ability to pool resources is true for private or public clouds. Another value of cloud computing is the ability to be more agile. Instead of having many independent and often underused servers deployed for your enterprise applications, pools of resources are aggregated, consolidated , and designed to be elastic enough to scale with the needs of your organization.

Cloud computing is not a location but rather a pool of resources that can be rapidly provisioned in an automated, on-demand manner. Read the quote below for the definition of cloud computing according to the U.S. National Institute of Standards and Technology.

Cloud Security Responsibilities :

In general terms, the cloud provider is responsible for security of the cloud, including the In security of the cloud data centers, and foundational networking, storage, compute, and virtualization services. The cloud customer is responsible for security in the cloud, which is further delineated by the cloud service model physical.

Service Models, Deployment Models, and Responsibilities

Visualization is a critical component of a cloud computing architecture that, when combined with software orchestration and management tools that are covered in this course, allows you to integrate disparate processes so that they can be automated, easily replicated, and offered on an as-needed basis.

Cloud Computing Service Models :

As data center managers face a burgeoning population of mobile users, the distributed workforce – with multiple endpoints and cloud applications – is forcing organizations to evolve both their in-house and cloud cybersecurity infrastructures.

Four Cloud Deployment Models:

Public Cloud:

Public cloud is a cloud infrastructure that is open to use by the general public. It's owned, managed, and operated by a third party (or parties), and it exists on the cloud provider's premises. Examples of public CSPs are Amazon Web Services (AWS), Google Cloud, and Microsoft Azure.

Community Cloud:

Community cloud is a cloud infrastructure that is used exclusively by a specific group of organizations.

Private Cloud:

Private cloud is a cloud infrastructure that is used exclusively by a single organization. It may be owned, managed, and operated by the organization or a third party (or a combination of both), and it may exist on-premises or off-premises.

Hybrid Cloud:

Hybrid cloud is a cloud infrastructure that comprises two or more of these deployment models and is, therefore, the best of both worlds: private data center for static, older workloads and public cloud for newer apps, agility, and scalability.

Consistent Security:

The same levels of application control and threat prevention should be used to protect both your cloud computing environment and your physical network. First, you need to be able to confirm the identity of your applications, validating their identity and forcing them to use only their standard ports. You also need to be able to block the use of rogue applications while simultaneously looking for and blocking misconfigured applications. Finally, application-specific threat prevention policies should be applied to block both known and unknown malware from moving into and across your network and cloud environment.

4.2 Cloud Native Technologies

A useful way to think of cloud native technologies is as a continuum spanning from virtual machines (VMs) to containers to serverless.

On one end are traditional VMs operated as stateful entities, as we've done for over a decade now. On the other are completely stateless, serverless apps that are effectively just bundles of app code without any packaged accompanying operating system (OS) dependencies

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Hypervison

Hypervisor software allows multiple, virtual guest operating systems to run concurrently on a single physical host computer. The hypervisor functions between the computer operating system and the hardware kernel.

Containers as a Service

As containers grew in popularity and use diversified, orchestrators such as Kubernetes (and its derivatives such as OpenShift), Mesos, and Docker Swarm became increasingly important to deploy and operate containers at scale. Containers-as-a-service (CaaS) platforms manage the underlying compute, storage, and network hardware by default and, although assembled from many more generic components, are highly optimized for container workloads. Click the tabs for more information about why orchestrators such as Kubernetes (and its derivatives such as OpenShift), Mesos, and Docker Swarm are difficult to operate at scale.

Complex to Set Up and Maintain :

Although these orchestrators abstract much of the complexity required to deploy and operate large numbers of microservices comprising many containers and running across many hosts, they can be complex to set up and maintain.

Difficult to Manage :

These orchestrators are focused on the container runtime and do little to assist with the deployment and management of underlying hosts. Although sophisticated organizations often use technologies such as thin VMs wrapped in automation tooling to address this, even these approaches do not fully unburden the organization from managing the underlying compute, storage, and network hardware.

Containers :

A container is a package of software that allows applications to run independently within a host operating system.

Container Orchestration :

Kubernetes Containers is an open-source orchestration platform that provides an application programming interface (AP) that enables developers to define container infrastructure in a declarative fashion, that is, infrastructure as code (IaC). Click the tabs for more information about application development using containers and microservices.

Traditional Security Protections :

(Firewall, Web Application Firewall (WAF), Intrusion Prevention System (IPS)/Intrusion Detection System (IDS))

Since organizations that use serverless architectures do not have access to the physical (or virtual) server or its operating system, they cannot deploy traditional security layers, such as endpoint protection, host-based intrusion prevention, WAFs, and so forth. Additionally, existing detection logic and rules have yet to be "translated" to support serverless environments.

4.3 Cloud Native Security

The Four Cs of Cloud Native Security:

The CNCF defines a container security model for Kubernetes in the context of cloud native security. Each layer provides a security foundation for the next layer.

Cloud:

The cloud (and data centers) provide the trusted computing base for a Kubernetes cluster. If the cluster is built on a foundation that is inherently vulnerable or configured with poor security controls, then the other layers cannot be properly secured.

Containers:

Securing the container layer includes container vulnerability scanning and OS dependency scanning, container image signing and enforcement, and implementing least privilege access.

code:

The application code itself must be secured. Security best practices for securing code include requiring TLS for access, limiting communication port ranges, scanning third-party libraries for known security vulnerabilities, and performing static and dynamic code analysis.

Prioritizing Software Security in the Cloud

The customer is ultimately responsible for providing security for the data, hosts, containers, and serverless instances in the cloud.

Public cloud service providers have done a great job with the build, maintenance, and updating of computing hardware, virtual machines, data storage, and databases along with the minimum baseline security protection mechanisms. However, the customer is ultimately responsible for providing security for the data, hosts, containers, and serverless instances in the cloud. Customers should follow three DevOps models and processes to better secure their data in the cloud.

DevOps Software Development Model

The DevOps software development model is enhancing or replacing the traditional software development life cycle (SDLC) model.

In the traditional software development model, developers write large amounts of code for new features, products, bug fixes, and such, and then pass their work to the operations team for deployment, usually via an automated ticketing system. The operations team receives this request in its queue, tests the code, and gets it ready for production – a process that can take days, weeks, or even months.

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Although automation is very important for DevOps culture, automation alone does not define DevOps.

The Importance of Security Standards

To prevent successful attacks, you must ensure your cloud resources and SaaS applications are correctly configured and adhere to your organization's security standards from day one. You also must make sure these applications and the data they collect and store are properly protected and

compliant to avoid costly fines, brand reputation damage, and loss of customer trust. Most if not all enterprise security teams must meet security standards and maintain compliant environments at scale and across SaaS applications.

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4.4 Hybrid Data Center Security

Data centers are rapidly evolving from a traditional, closed environment with static, hardwarebased computing resources to an environment in which traditional and cloud computing technologies are mixed.

The Hybrid Cloud:

Many organizations are using public cloud compute resources to expand private cloud capacity rather than expand compute capacity in an on-premises private cloud data center.

Virtualization of Guest Operating Systems:

The use of private cloud and public cloud compute resources to expand services is called the hybrid cloud. The virtualization of guest operating systems has forced organizations to adopt a hybrid cloud model over a traditional data center.

Private Cloud Traffic Types and Compute Clusters:

Organizations usually implement security to protect traffic flowing north-south, but this approach is insufficient for protecting east-west traffic within a private cloud. To improve their security posture, enterprises must protect against threats across the entire network, both northsouth and east-west.

Virtual Data Center Design:

In a virtual data center (private cloud), there are two different types of traffic, each of which is secured in a different manner: north-south and east-west.

The compute cluster is the building block for hosting the application infrastructure and provides the necessary resources in terms of compute, storage, networking, and security.

Private Cloud Security:

The use of virtual firewalls for east-west protection provides unprecedented traffic and threat visibility.

East-West Protection Benefits:

Virtual data center security best practices dictate a combination of north-south and east-west protection. Click the tabs for more information about the benefits that east-west protection provides.

Authorizes Allowed Apps:

Reduces Lateral Threats

Stops Threats ,protects from Data Theft

Traffic and Threat Visibility

An added benefit of using virtual firewalls for east-west protection is the unprecedented traffic and threat visibility that the virtualized security device can now provide. After Traffic logs and Threat logs are turned on, VM-to-VM communications and malicious attacks become visible. This virtual data-center awareness allows security teams to optimize policies and enforce cyberthreat protection (for example, IPS, anti-malware, file blocking, data filtering, and DoS protection) where needed.

Hybrid Cloud Security Evolution:

Security in a hybrid cloud evolves incrementally in four phases.

Phased Approach

The following approach to security in the evolving data center – from traditional three-tier architectures to virtual data centers and to the cloud – aligns with practical realities, such as the need to leverage existing best practices and technology investments, and the likelihood that most organizations will transform their data centers incrementally. This approach consists of four phases

5.Correlation between the summer industry Internship-I and the program outcomes(pos), Program specific Outcomes(psos)

Program outcomes are the knowledge, skills, and abilities that a program aims to develop in its students. In the context of a cybersecurity program, the outcomes might include understanding various security concepts, being able to analyze and mitigate security risks, and having the skills to protect systems and data from cyber threats.

Now, when it comes to cybersecurity, it's all about protecting computers, networks, and data from unauthorized access or damage. So, the program outcomes of a cybersecurity

program would align with the skills and knowledge needed to excel in the field. This could include understanding different types of cyber threats, knowing how to implement security measures, and being able to respond to incidents effectively.

Program-specific outcomes are more focused on the unique goals of a particular program. For example, a program might have specific outcomes related to ethical hacking, secure coding practices, or digital forensics. These outcomes would be tailored to the specific needs and objectives of the program.

In summary, the correlation between cybersecurity and program outcomes is strong because the program is designed to equip students with the necessary skills and knowledge to succeed in the cybersecurity field. The program-specific outcomes further refine this focus by aligning with the specific goals of the program.

6.Domain and Nature of Training carried out in Internship

In a cybersecurity internship, the training typically focuses on various domains and aspects of cybersecurity. The nature of the training can vary depending on the organization and the specific internship program. However, some common domains and areas of training in cybersecurity internships include:

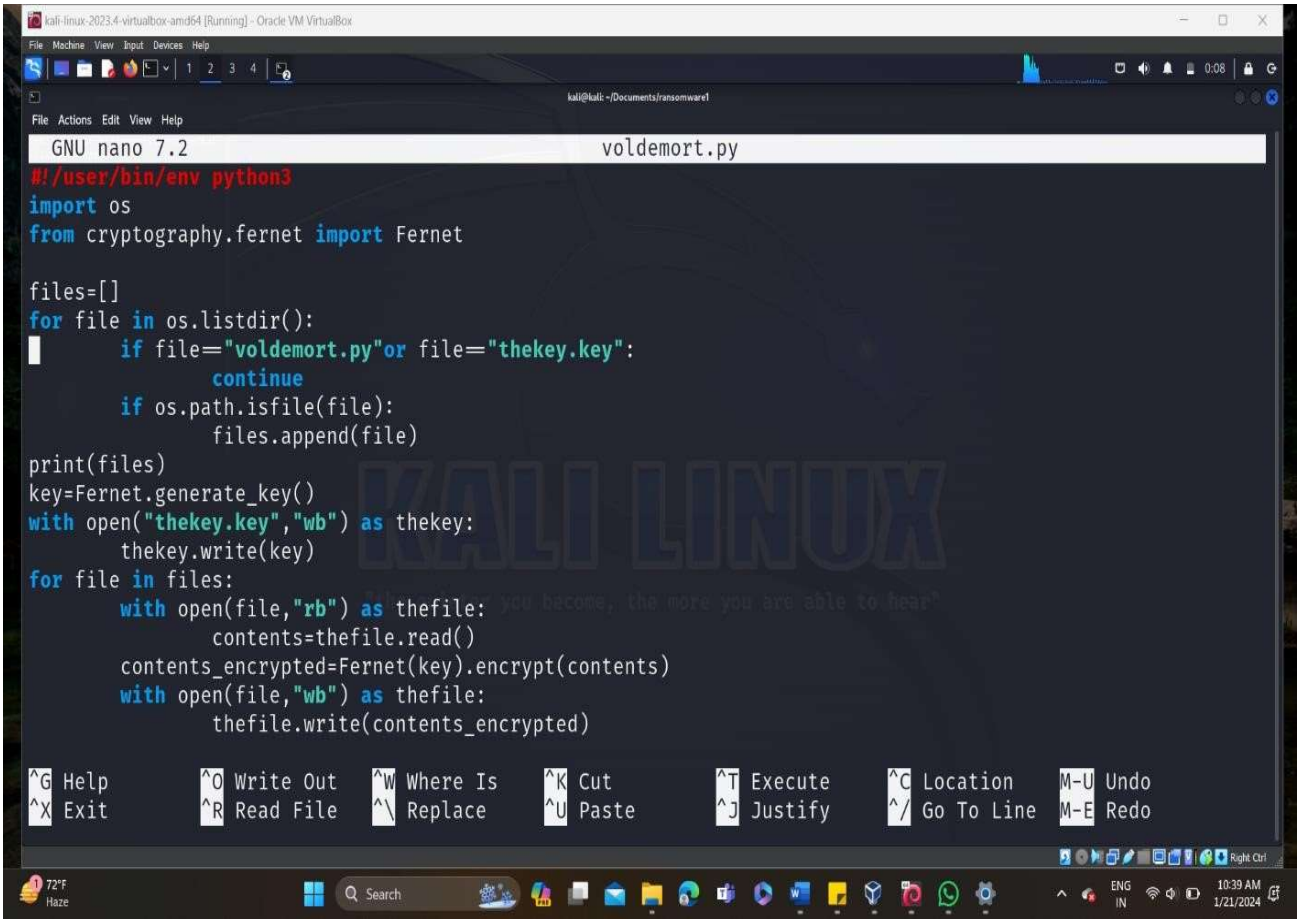
1. Network Security: This involves learning about securing computer networks, identifying vulnerabilities, implementing firewalls, and protecting against unauthorized access.
2. Incident Response: Interns may receive training on how to detect and respond to security incidents, including investigating security breaches, analyzing malware, and implementing incident response plans.
3. Vulnerability Assessment: This training helps interns learn how to identify and assess vulnerabilities in systems and networks, conduct security audits, and recommend remediation measures.
4. Security Operations: Interns may gain hands-on experience in security operations centers (SOCs), learning about monitoring and analyzing security events, managing security incidents, and implementing security controls.
5. Security Policy and Compliance: Training in this area focuses on understanding security policies, regulations, and compliance frameworks, and ensuring that organizations adhere to them.
6. Ethical Hacking: Some internships may provide training in ethical hacking, where interns learn how to identify and exploit vulnerabilities in a controlled environment to help organizations strengthen their security defenses.

These are just a few examples, and the specific training provided in a cybersecurity internship can vary. It's important to note that internships often provide a combination of theoretical knowledge and practical hands-on experience to develop the skills needed in the field of cybersecurity.

7.Project on Network Security:

Data encryption

Data encryption is like turning your information into a secret code that can only be understood by someone who has the right "key" to unlock it. Imagine you have a message, and you want to make sure that only the person you're sending it to can understand it. Instead of sending the message as it is, you use a special code to scramble the information. This code is created using a secret key.



DATA ENCRYPTION

Data Decryption:

Data decryption is like taking a secret coded message and using a special key to turn it back into its original, understandable form. Imagine you received a message that looks like a bunch of random letters and numbers – that's the encrypted data. To make sense of it, you need a specific key that matches the one used to encrypt the message.

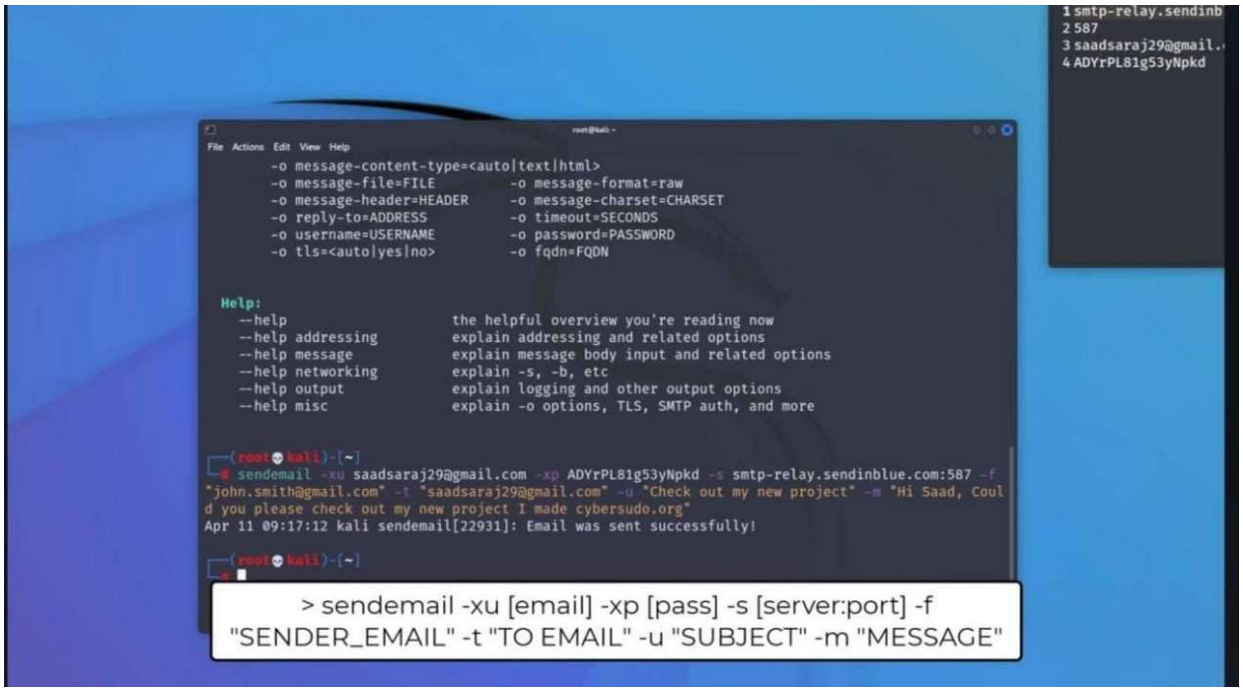
```
GNU nano 7.2                                   decrypt.py
#!/user/bin/env python3
import os
from cryptography.fernet import Fernet

files=[]
for file in os.listdir():
        if file=="voldemort.py"or file=="thekey.key" or file=="decrypt.py":
                continue
        if os.path.isfile(file):
                files.append(file)
print(files)
with open("thekey.key","rb") as key:
        secretkey=key.read()
for file in files:
        with open(file,"rb") as thefile:
                contents=thefile.read()
        contents_decrypted=Fernet(secretkey).decrypt(contents)
        with open(file,"wb") as thefile:
                thefile.write(contents_decrypted)

                                        [ Read 19 lines ]
^G Help     ^O Write Out   ^W Where Is    ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit     ^R Read File   ^\ Replace     ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

DATA DECRYPTION

```
zsh: corrupt history file /home/kal
i/.zsh_history

┌──(kali㉿kali)-[~]
└─$ cd Documents

┌──(kali㉿kali)-[~/Documents]
└─$ cd ransomware1

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ ls
decrypt.py  filer.txt  file.txt  hey.txt  pleasedonthurtme.txt  thekey.key  voldemort.py  voldemort.py.save

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ cat file.txt
this is a file

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ python3 voldemort.py
['voldemort.py.save', 'pleasedonthurtme.txt', 'filer.txt', 'file.txt', 'hey.txt']

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ cat file.txt
gAAAAABlrRiXJQWtG8wDcF7n2uUP06z7g6GMykIrNibjPPIF7pkHEKH9WFlGcgADMR0×9hUd73Tj-CMPXz17XDuoYZpO8JG68w==

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ python3 decrypt.py
['voldemort.py.save', 'pleasedonthurtme.txt', 'filer.txt', 'file.txt', 'hey.txt']

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$ cat file.txt
this is a file

┌──(kali㉿kali)-[~/Documents/ransomware1]
└─$
```

Email spoofing:

Email spoofing is a deceptive technique used by malicious actors to forge the sender's address in an email to make it appear as if it's coming from a trusted source. Essentially, it involves manipulating the email header information to trick the recipient into believing the message is legitimate



EMAIL SPOOFING

Password cracker:

a password cracker is a tool designed to break or crack passwords, often through various techniques such as brute-force attacks or dictionary attacks. The latter is used with malicious intent to gain unauthorized access to an account or system by systematically attempting different passwords until the correct one is found. In summary, a password generator enhances security by creating robust passwords, while a password cracker poses a security threat by attempting to compromise passwords without authorization

```
import random import pyautogui
chars="abcdefghijklmnopqrstuvwxyz0123456789"
allchar=list(chars)
pwd=pyautogui.password("enter a password")
sample_pwd="" while(sample_pwd!=pwd):
    sample_pwd=random.choices(allchar,k=len(pwd))
print("<====="+str(sample_pwd)+"====>")
if(sample_pwd==list(pwd)):
        print("the password is:"+"".join(sample_pwd))
break
```

PASSWORD CRACKER

References:

www.springboard.com www.knowledgehut.com
www.securityscoreguard.com
www.stackoverflow.com